

# Comparing Routes to Complex Adaptation

*Conditional Waiting-Time Models, Horizontal Transfer, and State-First Hypotheses*

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## Abstract

Complex traits can arise through many combinations of mutation, standing variation, recombination, sequential fixation, stochastic tunneling, gene and genome duplication, horizontal transfer, developmental plasticity, and state-dependent cellular processes. An earlier version of this paper framed three routes as a trilemma and treated a waiting time proportional to  $p_b^{-L}$  as a hard result. That scaling is valid only for a restrictive toy model in which all  $L$  required changes occur as one prescribed independent event with constant probability. It is not a general theorem of evolutionary dynamics. This correction recasts the paper as a conditional route-comparison framework. It places the simultaneous-route calculation beside explicit population- genetic alternatives, treats horizontal transfer as an empirically real but case-specific mechanism, and defines computed state-first assembly (COT) as an unconfirmed order-of-events hypothesis. A seeded Wright-Fisher illustration shows how allowing inherited intermediates and multiple mutation orders changes waiting times and makes them depend on population size, mutation supply, fitness landscape, and the chosen endpoint. A target-set formulation, explicit censoring likelihood, nonexclusive route mixture, and randomized COT contrast make the inferential requirements auditable. The simulation is not an estimate for any real trait. The paper's durable contribution is a model and evidence discipline: route hypotheses must be nonexclusive, parameterized, and linked to distinct observables, interventions, and falsifiers.

## Scope and evidence boundary

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**Source lineage:** successor to Zenodo record 21234562, concept DOI 10.5281/zenodo.21234561

**Release scope:** a methods-focused correction and route-comparison framework. Any trait-specific biological application requires a separately parameterized model, additional sources, and specialist review.

## 1. Correction of the Central Claim

Suppose one prescribed route requires  $L$  independent changes, each with probability  $p_b$ , and all changes must occur in the same trial. Then the event probability is  $p_b^L$ . With  $N$  independent trials per generation, the event probability per generation is

$$q = 1 - \left(1 - p_b^L\right)^N. \quad (1)$$

If  $T$  is the number of generations through the first event, the geometric waiting time is

$$\mathbb{E}[T] = \frac{1}{q}. \quad (2)$$

When  $Np_b^L \ll 1$ , the first-order rare-event approximation is

$$\mathbb{E}[T] \approx \frac{1}{Np_b^L}. \quad (3)$$

These equations are correct for the stated simultaneous prescribed-route model.

The derivation does not apply unchanged when intermediate states are inherited, selected, lost, recombined, or reached in multiple orders. Nor does it include standing variation, stochastic tunneling, changing environments, epistasis, population structure, or migration. The earlier wording that established a hard boundary regardless of offspring number or selection strength is withdrawn.

## 2. Route Families Are Not an Exhaustive Trilemma

This paper uses three broad families for organizing hypotheses, not for forcing an exhaustive choice.

### 2.1 Conventional population-genetic assembly

This family includes sequential fixation, segregating intermediates, standing variation, recombination, tunneling across low-fitness intermediates, multiple genetic routes, and lineage-specific developmental constraints. Its models must state population size, mutation and recombination rates, dominance, selection, epistasis, demography, and the endpoint definition.

### 2.2 Horizontal transfer

HGT is an empirically established mechanism in particular lineages and traits. Candidate cases require donor or mobile-element evidence, phylogenetic discordance, integration history, and a comparison against vertical and convergent alternatives. Gene duplication and transposition are not HGT: they are within-lineage processes, even though all three can be discussed at a higher abstraction level as module reuse.

### 2.3 Computed state-first assembly

COT is defined here as a proposed order of events: an adaptive cellular state is learned or stabilized before the relevant heritable genomic change, and the state causally changes the distribution of later genomic outcomes. This is not established by cellular decision-making alone. It is also not synonymous with the Baldwin effect, plasticity-first evolution, or genetic assimilation.

These families can coexist. A transferred module can be refined by ordinary selection; plasticity can expose new selective regimes; and different parts of one trait can have different histories.

## 3. Population-Genetic Replacement Framework

For each candidate trait, define a route model

$M = \{\text{population process, genotype / state space, transition rules, fitness landscape, demography, observation model, endpoint}\}.$

The question is not whether a generic path length is large. The question is whether a specified process reaches a specified endpoint with a probability and time distribution compatible with the lineage, population, and data.

For route model  $r$ , let  $X_t^{(r)}$  be its state at generation  $t$ , and let  $A$  be the declared target set. The route-specific first-arrival time is

$$T_A^{(r)} = \inf \left\{ t \geq 0 : X_t^{(r)} \in A \right\}. \quad (4)$$

This definition forces the endpoint to be declared. First appearance, establishment, fixation, and functional expression correspond to different target sets and therefore different waiting-time questions.

Primary population-genetic work demonstrates why this matters. Lynch and Abegg analyze establishment of complex adaptations with sequential fixation and tunneling rather than requiring a complete route in one event. Durrett and Schmidt show that even two-change waiting times depend strongly on population size and model structure. Those results do not prove that every conventional route is fast; they block treating  $p_b^{-L}$  as a universal impossibility law.

Required sensitivity dimensions include:

- effective population size and reproductive variance;
- mutation supply and target size;
- intermediate and final fitness;
- number and connectivity of viable paths;
- standing variation and recombination;
- environmental change and population structure; and
- whether the endpoint is first appearance, establishment, fixation, or functional expression.

## 4. Reproducible Toy Comparison

The accompanying script compares two deliberately different models.

1. The simultaneous prescribed-route calculation requires all loci to change in one generation in one individual.
2. A forward-only haploid Wright-Fisher illustration allows intermediate genotypes to be inherited and all mutation orders to contribute.

The simulation uses fixed, case-specific seeds and three illustrative fitness landscapes: beneficial intermediates, neutral intermediates, and a deleterious valley. It records first appearance of the full target and the generation at which the target reaches a declared population frequency. Four additional cases vary mutation rate, population size, and locus count. All inputs are in `model/parameters.json` and generated results are in `model/results.json`. Censored replicates are counted explicitly; medians and quantiles are null if no replicate reaches the endpoint.

This comparison has three narrow purposes:

- demonstrate that the whole-route exponential is tied to one transition model;
- expose dependence on population, mutation, fitness, and endpoint; and
- provide a reproducible sensitivity surface that can later be replaced by a trait-specific model.

With the committed illustrative inputs ( $N=100$ ,  $L=3$ , per-locus forward mutation rate 0.02, 80 replicates), the simultaneous prescribed-route model has an expected first-event wait of about 1,250.5 generations. The sequential simulation produced the following seeded summaries:

| Fitness landscape     | Median first target | Median generation at 90% target frequency |
|-----------------------|---------------------|-------------------------------------------|
| Beneficial steps      | 11.0                | 51.0                                      |
| Neutral intermediates | 11.0                | 47.5                                      |
| Deleterious valley    | 12.0                | 43.0                                      |

All replicates completed before the 2,500-generation cap. These values are not ordered estimates of biological difficulty. In this small forward-only model, the final genotype is strongly favored, mutations do not reverse, and the endpoint is a population-frequency threshold. The table demonstrates model dependence, not a universal advantage for any landscape.

The four seeded sensitivity cases give:

| Case                                 | Simultaneous expected wait | Median first target | Median 90% threshold |
|--------------------------------------|----------------------------|---------------------|----------------------|
| Neutral L=3, mutation<br>0.01, N=100 | 10,000.5                   | 21.5                | 64.0                 |
| Neutral L=3, mutation<br>0.02, N=50  | 2,500.5                    | 16.5                | 49.0                 |
| Neutral L=3, mutation<br>0.02, N=200 | 625.5                      | 9.0                 | 46.0                 |
| Neutral L=4, mutation<br>0.02, N=100 | 62,500.5                   | 18.0                | 56.0                 |

Each case has 40 replicates and all reached the threshold before its declared cap. Halving mutation rate, changing population size, or adding a locus changes both the simultaneous formula and the sequential distribution. These four cases are a regression matrix, not a complete sensitivity analysis. Their medians do not support monotonic biological ranking: the model omits reverse mutation, recombination, dominance, changing demography, and population structure, and it uses a strongly favored final genotype.

It is not calibrated to a real organism or trait. Its forward-only mutation, haploid reproduction, simple fitness-by-derived-count landscape, and lack of recombination are strong simplifications. No HGT or COT inference follows from its output.

## 5. Conditional Identifiability

Evidence erosion can make routes harder to distinguish, but this is not a route-independent theorem. Let  $G_r(\theta)$  be a generative model for route  $r$ , and let  $O_\phi$  be an observation process that includes deletion, substitution, incomplete sampling, and measurement error. Identifiability is a property of the induced distributions

$$p(D \mid r, \theta, \phi), \quad (5)$$

not of route labels alone. Formally, routes  $r_1$  and  $r_2$  are observationally nonidentifiable when there exist admitted parameter choices  $\theta_1$  and  $\theta_2$  for which

$$p(D \mid r_1, \theta_1, \phi) = p(D \mid r_2, \theta_2, \phi) \quad \text{for every admitted observable } D. \quad (6)$$

If they differ on a measured statistic with adequate power, they may instead be distinguishable.

## 5.1 Censored waiting-time data

Finite experiments and simulations often end before a route reaches its target. Let  $C_i$  be the observation limit for replicate  $i$ . The recorded pair must distinguish an observed arrival from censoring:

$$\begin{aligned} Y_i &= \min(T_{A,i}^{(r)}, C_i), \\ \delta_i &= \mathbf{1}[T_{A,i}^{(r)} \leq C_i]. \end{aligned} \quad (7)$$

If route  $r$  has arrival density or mass function  $f_r$  and survival function  $S_r$ , a standard right-censored contribution is

$$\mathcal{L}_r = \prod_i f_r(Y_i)^{\delta_i} S_r(Y_i)^{1-\delta_i}. \quad (8)$$

Replacing censored observations with the cap, discarding them, or reporting a median only among completed replicates can bias a route comparison. The current toy fixtures report censoring transparently but do not estimate this likelihood; that remains a release gate for any empirical application.

## 5.2 Nonexclusive route mixtures

Because the route families can operate together, a dataset need not have a single route label. One explicit representation is

$$\begin{aligned} p(D \mid \Theta, \phi) &= \sum_{r=1}^R \pi_r p(D \mid r, \theta_r, \phi), \\ \pi_r &\geq 0, \quad \sum_{r=1}^R \pi_r = 1. \end{aligned} \quad (9)$$

The weights are model components, not proof that a historical event decomposed in exactly those proportions. A mixture can also be nonidentifiable when different weights and parameters induce the same observable distribution. Accordingly, the paper does not convert a poor fit for one conventional model into evidence for HGT or COT; it requires positive, route-specific observables.

A defensible analysis must therefore state:

1. the route-specific generative models;
2. the preservation and sampling process;
3. candidate distinguishing statistics;
4. priors or parameter bounds where used;
5. calibration and posterior-predictive or likelihood checks; and
6. regions where the data cannot distinguish routes.

Erosion alone does not favor HGT over convergence. It may reduce evidence for both.

## 6. HGT Evidence Standard

The existence of well-supported HGT cases establishes feasibility, not necessity. A route-specific HGT claim should combine several independent signals where possible:

- gene-tree / species-tree discordance with adequate taxon sampling;
- synteny, compositional, or integration signatures;
- a plausible donor and timing relation;
- mobile-element or vector evidence;
- exclusion of contamination and assembly artifacts; and
- explicit comparison with duplication, loss, incomplete lineage sorting, and convergent sequence evolution.

Ancient transfer may become difficult to identify as these signals decay. That uncertainty should be reported as uncertainty, not converted into a general claim that HGT must have occurred.

## 7. COT as an Order-of-Events Hypothesis

COT's strongest testable form is temporal and causal. A study would need to show:

1. a reproducible adaptive cellular state before the candidate genomic change;
2. operational evidence that the state was acquired rather than genetically pre-specified;
3. a later heritable genomic change associated with stabilization;
4. a changed distribution of genomic outcomes conditional on the prior state;
5. loss or alteration of that genomic bias when the state-forming process is blocked; and
6. replication across independent populations with sequencing and lineage controls.

For a randomized state-induction assignment  $Z$  and a prespecified later genomic endpoint  $G_t$ , the primary directional contrast can be written as

$$\Delta_{\text{COT}}(t) = \Pr(G_t = 1 \mid Z = \text{induce}) - \Pr(G_t = 1 \mid Z = \text{control}). \quad (10)$$

A positive contrast is not sufficient by itself. The state must precede the genomic endpoint; blocking the state-forming process must reduce the contrast; lineage and sequencing controls must exclude pre-existing variants and sample mixing; and negative-control loci must test whether the intervention merely raises genome-wide mutation or selection pressure. These requirements turn COT from a retrospective story into a preregisterable intervention test.

Alternative explanations include ordinary selection on standing variants, stress-induced but nondirected mutagenesis, population sorting, cryptic genetic variation, and measurement timing. COT is weakened or falsified when genomic change consistently precedes the adaptive state, when state blockade leaves the genomic distribution unchanged, or when preregistered replication fails.

Kramer et al. experimentally show that cellular state can condition single-cell decision-making. Gunawardena reviews how learning concepts may be applied outside nervous systems. These sources motivate state-dependent experiments; neither demonstrates a learned state causing a later heritable genomic change.

The machine-readable route ledger in `data/route-predictions.jsonl` records the assumptions, observables, and falsifiers for all three families.

## 8. Version and Correction Record

Zenodo record 21234562 is the public predecessor of this correction. The present version changes the scientific endpoint from a general evolutionary trilemma to a methods-focused comparison of explicitly conditional route models. It preserves the motivating distinction among conventional assembly, horizontal

transfer, and computed state-first assembly while withdrawing the claim that  $p_b^{-L}$  supplies a general evolutionary waiting-time law.

The linked version should remain under concept DOI 10.5281/zenodo.21234561. The public description should identify this as a corrected methods preprint, state that the numerical example is illustrative rather than trait-calibrated, and link readers to the predecessor rather than silently replacing its record.

## 9. Scope and Future Validation

This release does not select or analyze a biological trait dataset. For a future empirical application, the population process, mutation and recombination model, selection regime, path structure, demography, observation model, and endpoint must be specified for that case. Such an application would also require trait-specific primary sources, sensitivity and identifiability analysis, and population-genetics and statistical review.

The COT component is a prospective order-of-events hypothesis. A future test requires a preregistered intervention, negative controls, time-resolved cell-state measurements, a heritable endpoint, and an analysis that can distinguish state-first effects from ordinary selection, stress response, plasticity, and measurement bias. Those future empirical requirements do not convert the present methods paper into an empirical claim.

## 10. Conclusion

The earlier trilemma overreached because it promoted a restrictive simultaneous route model into a general evolutionary boundary. The corrected framework is more useful: it asks which specified route models are feasible and distinguishable for a specified trait and dataset. Conventional evolution, HGT, and state-first processes are nonexclusive hypothesis families. HGT is real but case-specific; COT is testable but unconfirmed; and evidence erosion creates model-dependent uncertainty rather than a default winner. Target-set hitting times, censored-data likelihoods, route mixtures, and randomized contrasts make that program operational without presenting a toy model as biological proof. The resulting program is harder than a closed-form trilemma, but it is scientifically answerable.

## Declarations

- **Data availability:** No new biological dataset is analyzed. The numerical results are deterministic toy-model outputs used to demonstrate conditional route behavior.
- **Code availability:** The finite-state model, parameters, tests, and machine-readable route ledger accompany the release package.
- **Author contributions:** The paper is published in the established Blumberg-Golan coauthored lineage. The present correction formalizes and narrows that joint route-comparison program.
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- **Competing interests:** The authors declare no competing interests.

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